

Exploratory Data Analysis

Large-Scale Retail Transaction Dataset

100,000 transactions • 3 years • 86 statistical tests

By: Srishti Sharma



Objective

- ✓ Assess **data quality and structure** through comprehensive EDA
- ✓ Analyse **univariate, bivariate, and multivariate** relationships
- ✓ Detect and evaluate **outliers** and their impact on analysis
- ✓ Investigate **temporal trends, seasonality, and revenue stability**
- ✓ Evaluate **statistical and practical significance** using appropriate hypothesis tests and effect sizes
- ✓ Answer key **business questions** using aggregation and exploratory analysis
- ✓ Build baseline **regression and classification models** to assess predictive capability

Methodology



The Central Finding

86

*ANOVA · Chi-square ·
Correlation
VIF · Point-biserial · Trend
tests*

formal significance
tests run

**``final_price``
is the only variable with genuine structure**

It is mathematically derived from `product_price`, `quantity`, and `discount_percentage` — not the result of customer or market behavior.

Every other dimension tested — demographics, geography, product category, payment method, shipping type — is statistically indistinguishable from uniform/random.

This conclusion held after Benjamini-Hochberg multiple-testing correction, bootstrap confidence intervals, and a statistical power analysis confirming the tests could detect effects far smaller than practically meaningful.

Dataset at a Glance

1,00,000

Transactions

3 Years

01 Jan 2023 – 30 Dec 2025

₹639.79 Cr

Total Revenue

14.81%

Overall Return Rate

Data Quality Findings

- ✓ **Missing Values:** 0 across all 18 columns — dataset is complete
- ✓ **Duplicate Values:** 0 — every one of the 100,000 transactions is unique
- ✓ **Repeat Customers:** 0% — every customer_id appears exactly once
- ✓ **Data Coverage:** 2023-01-01 → 2025-12-30 (1,095 unique dates)
- ✓ **Data Types:** All numerical columns correctly typed as int64/float64; order_date parses cleanly to datetime64
- ✓ **Value Range:** No negative values; all within plausible bounds
- ✓ **Data Completeness:** No gaps — every calendar day in range has ≥1 order

	dtype	null_count	null_%	unique_vals
customer_id	object	0	0.00	100000
age	int64	0	0.00	48
gender	object	0	0.00	2
city	object	0	0.00	10
state	object	0	0.00	9
customer_segment	object	0	0.00	3
order_id	object	0	0.00	100000
order_date	datetime64[ns]	0	0.00	1095
product_category	object	0	0.00	6
product_subcategory	object	0	0.00	21
product_price	int64	0	0.00	48449
quantity	int64	0	0.00	4
discount_percentage	int64	0	0.00	31
final_price	float64	0	0.00	97092
payment_method	object	0	0.00	4
shipping_type	object	0	0.00	2
delivery_days	int64	0	0.00	10
return_status	object	0	0.00	2

Feature Engineering

Behavioural

customer_age_group -
Binned age segments

high_value_order -
Boolean flag for top
orders

revenue_per_unit -
final_price / quantity

Price & Discount

price_bucket -
Tiered price segments

discount_bucket -
Binned discount ranges

discount_impact -
Absolute savings amount

Temporal

year/ quarter /month -
Date decomposition

day of_week –
Weekday / weekend
pattern

Data loss -
All 100,000 rows retained

Understanding the Dataset

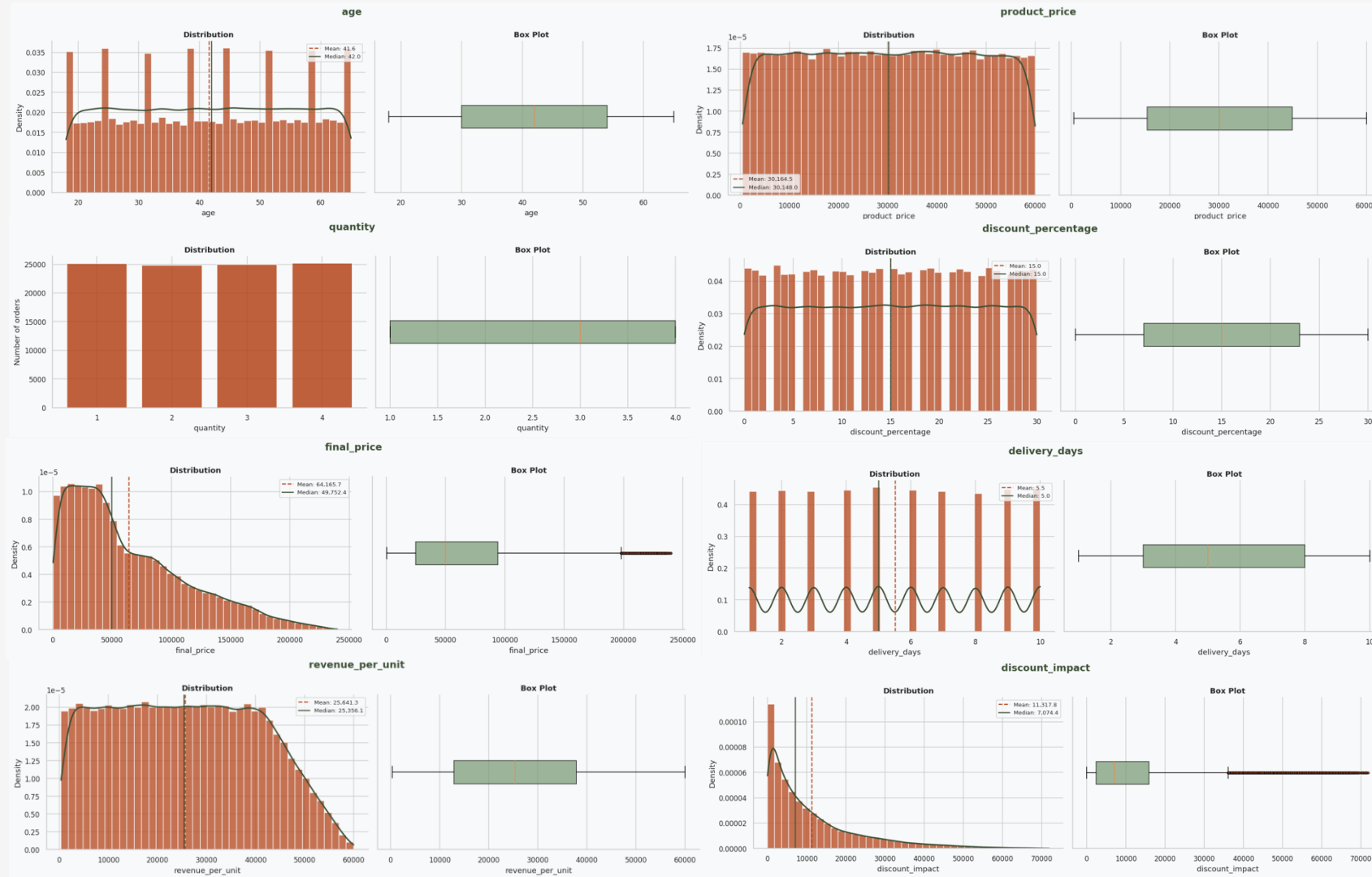
Statistical Findings

Variable	Min	Max	Mean	Median	Std. Dev.	Skewness	Kurtosis	IQR	Mode	CV %	Distribution	Outliers	Key Observation
age	18	65	41.56	42	13.85	-0.005	-1.2012	24	34	33.33%	Approximately symmetric	0	Near-uniform distribution across ages 18–65.
product_price	500	59,998.00	30,164.53	30,148.00	17,139.68	0.0032	-1.1951	29,582.00	41,274	56.82%	Approximately symmetric	0	Broad price range with an approximately uniform distribution.
quantity	1	4	2.5	3	1.12	-0.0031	-1.3646	3	4	44.77%	Approximately symmetric	0	Discrete values (1–4 units) with balanced frequencies.
discount_percentage	0	30	15.01	15	8.95	-0.0036	-1.2016	16	3	59.65%	Approximately symmetric	0	Discounts are evenly spread across the 0–30% range.
final_price	357	2,39,980.00	64,165.72	49,752.39	49,969.28	0.9347	0.169	69,341.07	9,959.04	77.88%	Moderately right-skewed	1,335 (1.34%)	Derived financial metric with structural positive skew and moderate outliers.
delivery_days	1	10	5.51	5	2.87	0.0023	-1.2215	5	5	52.18%	Approximately symmetric	0	Delivery times are evenly distributed from 1–10 days.
revenue_per_unit	357	59,995.00	25,641.29	25,356.06	14,903.12	0.1142	-1.0431	24,911.58	2,664	58.12%	Approximately symmetric	0	Derived metric remains nearly symmetric because dividing by quantity reduces overall skewness.
discount_impact	0	71,956.80	11,317.82	7,074.35	12,167.33	1.6702	2.8159	13,492.30	0	107.51%	Highly right-skewed	5,567 (5.57%)	Most variable metric; high-value purchases create a long right tail and the largest number of outliers.

Key Insight: The dataset is fully clean, complete, and continuous — no missing values, duplicates, or gaps in coverage. Base transaction fields are symmetric, but derived financial metrics (final_price, discount_impact) are right-skewed and highly variable, flagging them as priority areas for outlier review.

Univariate Analysis

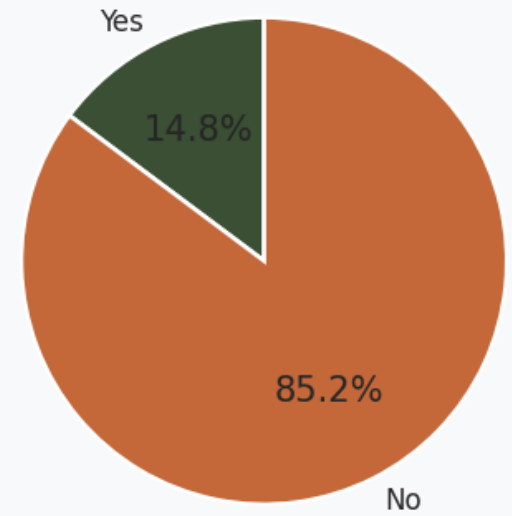
Numerical Analysis



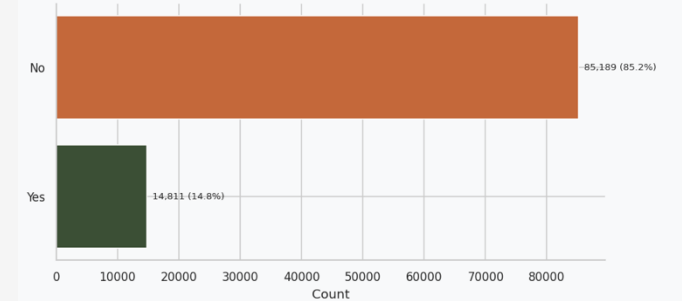
Categorical Analysis

return_status (2 categories)

Share of Total



Bar Chart



Univariate Analysis

Symmetry:

Six of eight numerical variables exhibit near-perfect symmetry ($|\text{skew}| < 0.01$) with no meaningful outliers. Only `final_price` and `discount_impact` show noticeable positive skewness due to their mathematical construction.

Representative Example:

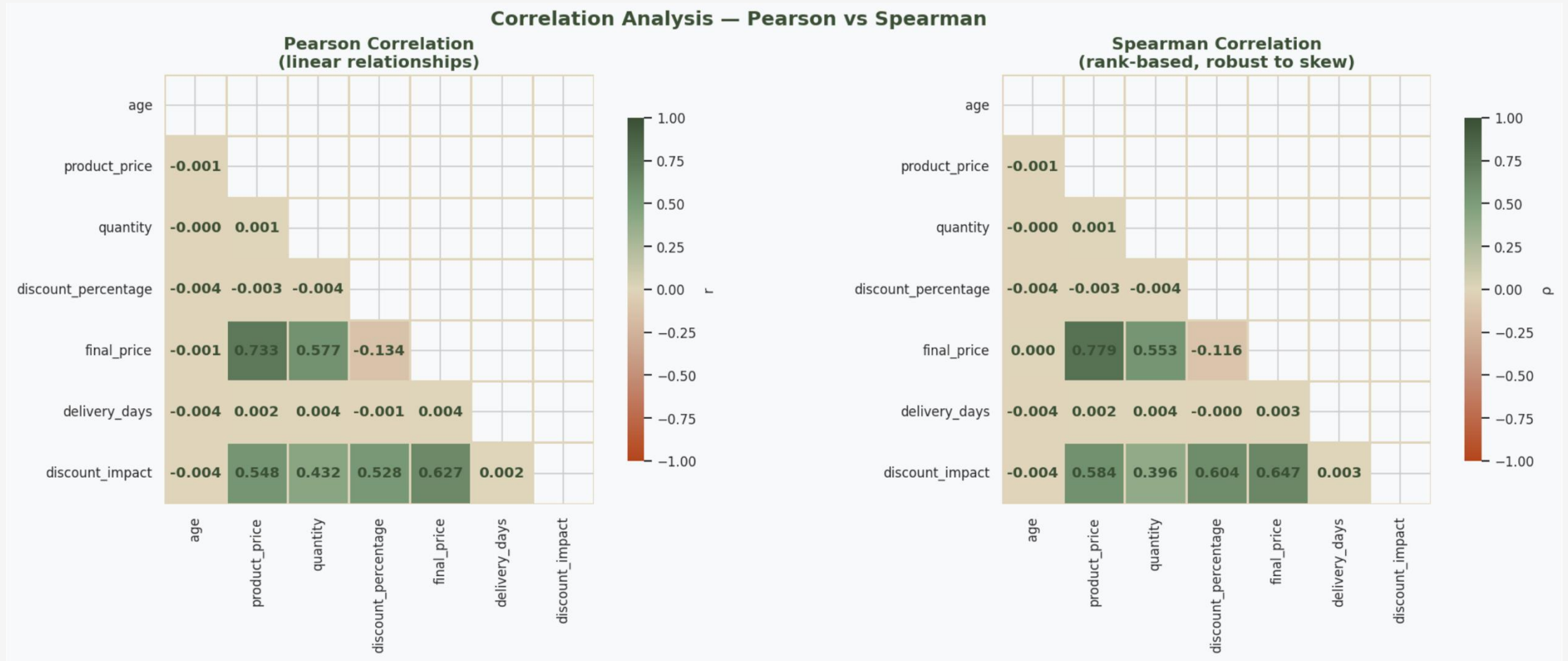
`final_price` and `discount_impact` are the only variables with meaningful skewness, reflecting the multiplicative pricing formula rather than genuine customer behaviour.

Categorical Balance:

Fifteen categorical variables were examined, with only `return_status` showing genuine imbalance (14.81% returned, 85.19% not returned). All other apparent imbalances arise from unequal category splits, engineered variable structure, or state-level aggregation artefacts.

Univariate analysis confirms a highly balanced synthetic dataset, with meaningful variation limited to derived pricing variables and `return_status`.

Correlation Structure — Proving the Central Finding



Every strong correlation traces back to final_price's own formula — visual proof that structure is mathematical, not behavioral.

Correlation Structure — Observations

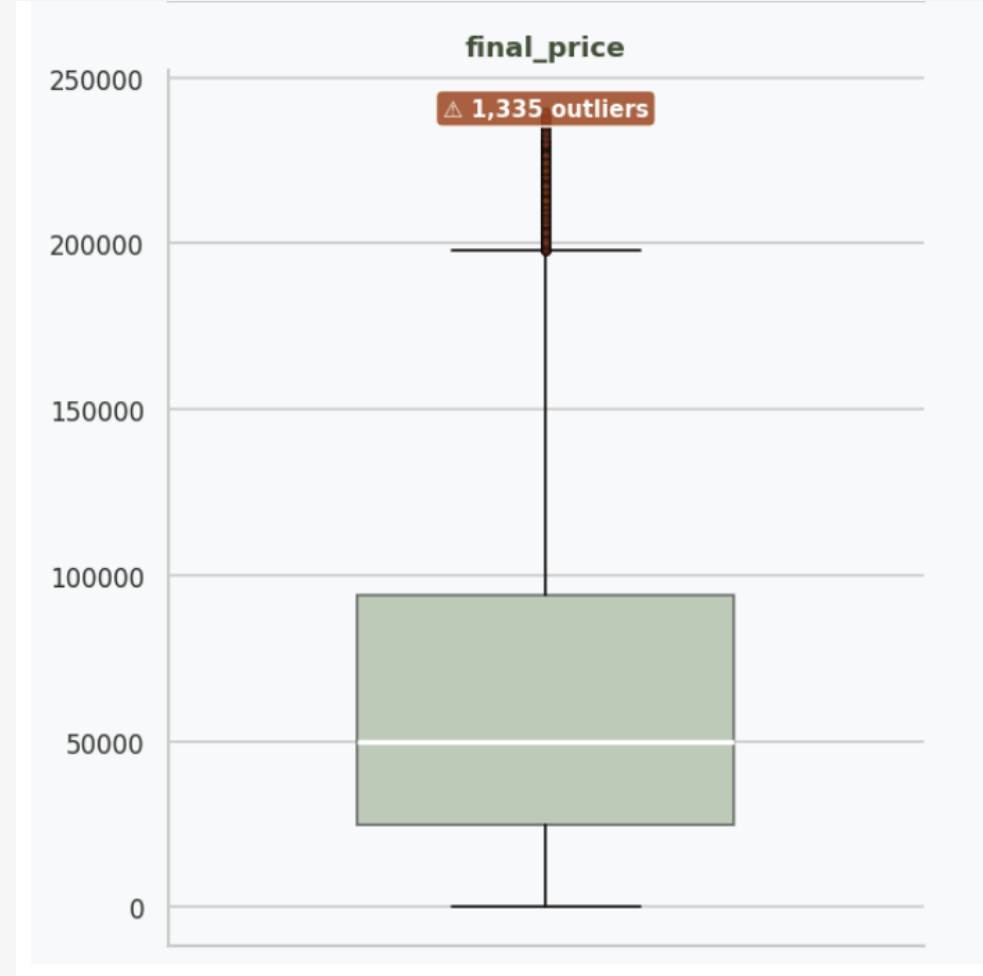
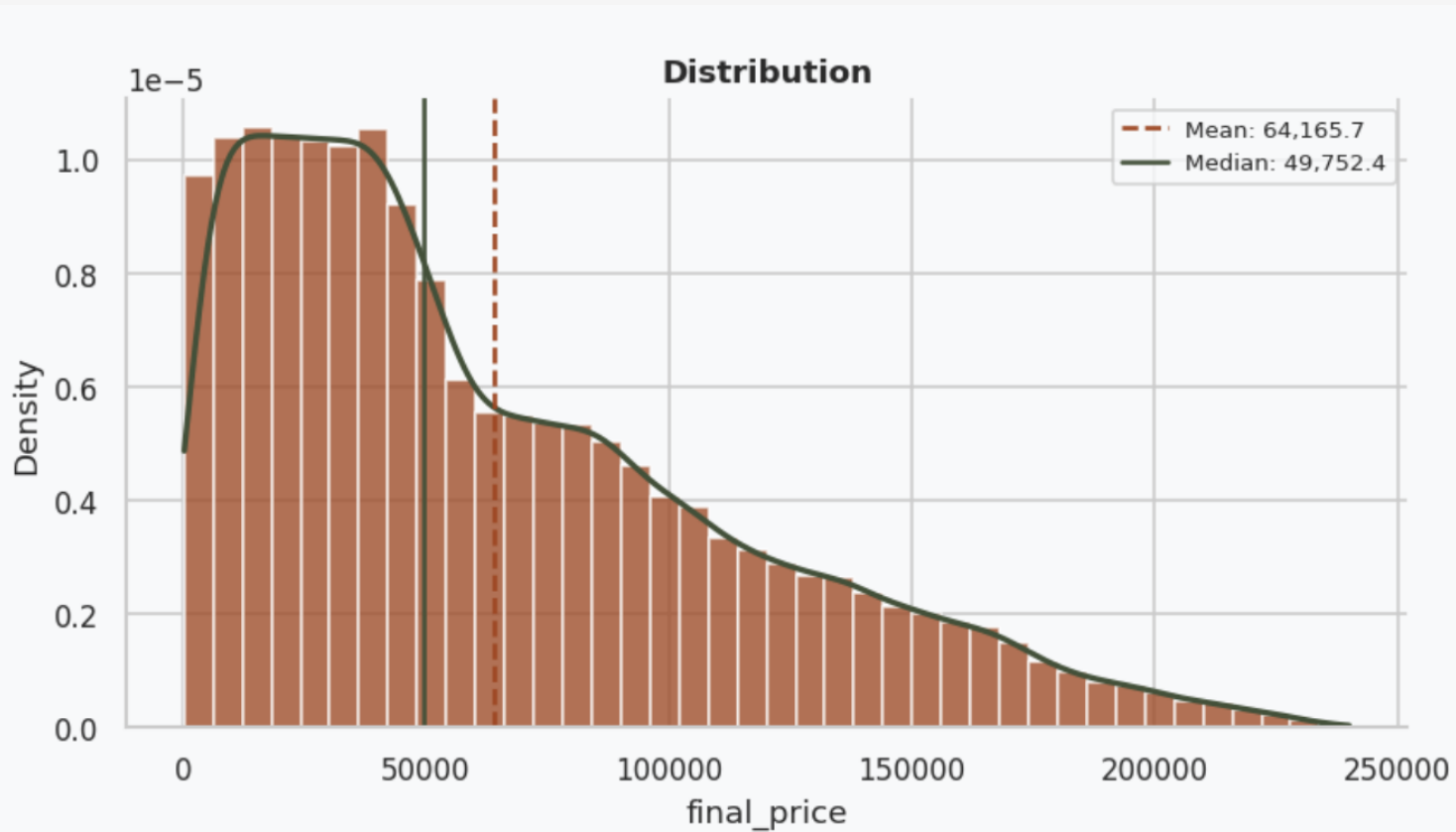
Pearson vs Spearman gap:

Compared Pearson and Spearman correlations to assess non-linearity. Only product_price showed a meaningful increase ($0.733 \rightarrow 0.779$; $\Delta = 0.046$), while quantity ($\Delta = 0.024$), discount_impact ($\Delta = 0.020$), and discount_percentage ($\Delta = 0.019$) showed only minor differences. No variable exceeded the 0.10 non-linearity threshold.

Correlation Ranking:

product_price was the strongest predictor of final_price, followed by discount_impact and quantity. discount_percentage contributed only weakly on its own because its effect is largely captured through the derived discount_impact variable.

Outlier Detection



Outliers were concentrated in two derived variables. Using the IQR method: final_price had 1,335 outliers (1.33%) and discount_impact had 5,567 outliers (5.57%). Alternative methods produced lower counts for final_price (Z-score 503, Modified Z-score 580) and varied counts for discount_impact (Z-score 1,931, Modified Z-score 5,640).

Outlier Treatment

Winsorization

Applied to final_price to reduce influence of extremes. Skewness reduced from 0.9347 to 0.8801 and IQR outliers were eliminated post-treatment.

No Treatment for discount_impact

Left unchanged because it is a derived variable; altering it would break mathematical relationships with product_price, quantity, and discount_percentage.

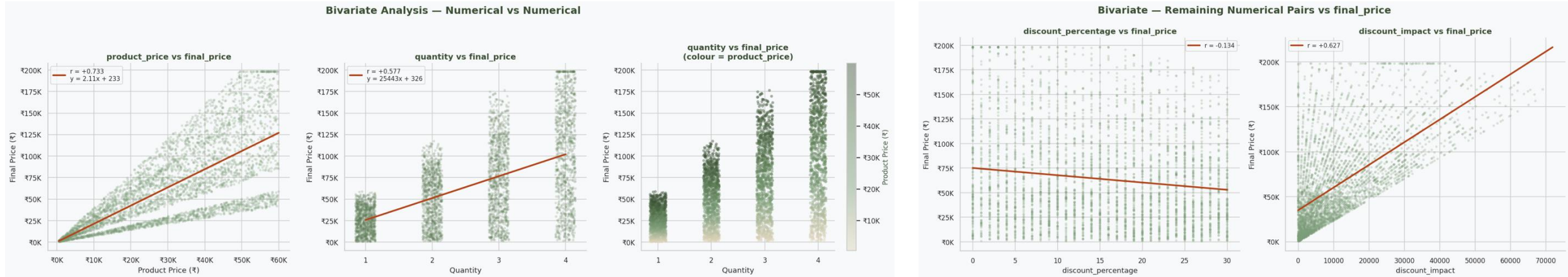
Sensitivity

Testing across multiple IQR thresholds showed conclusions remained stable, indicating outliers did not materially influence analyses.

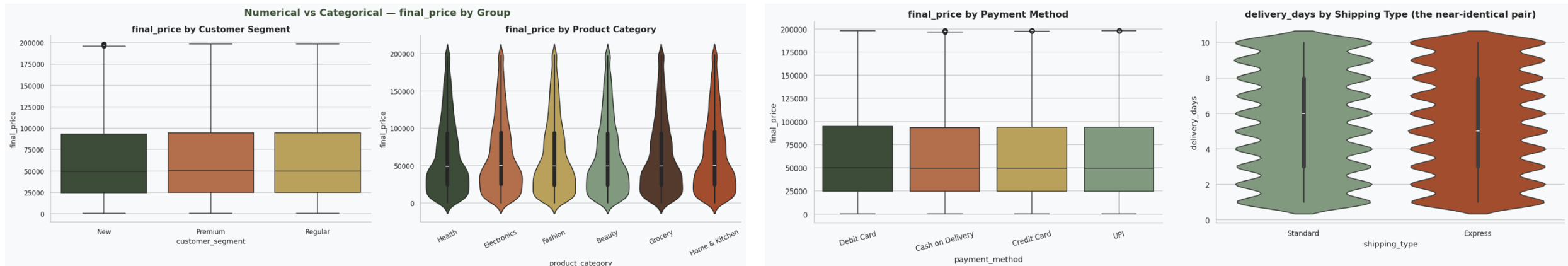
Outliers had minimal impact on overall findings. Winsorization reduced the influence of extreme final_price values while preserving the deterministic pricing relationships within the dataset.

Bivariate Analysis

Numerical Analysis vs Numerical Analysis



Numerical Analysis Vs Categorical Analysis



Bivariate Analysis

Correlation Strength:

Only pricing variables showed meaningful relationships with final_price: product_price ($r = +0.733$), discount_impact ($r = +0.627$), and quantity ($r = +0.577$). Demographics, geography, shipping, and delivery variables all showed $r \approx 0$, indicating negligible relationships with spending.

Suppression Effect:

The correlation between discount_percentage and final_price strengthened from $r = -0.134$ to $r = -0.194$ after controlling for product_price, revealing a suppression effect rather than a stronger direct relationship.

Statistical Significance:

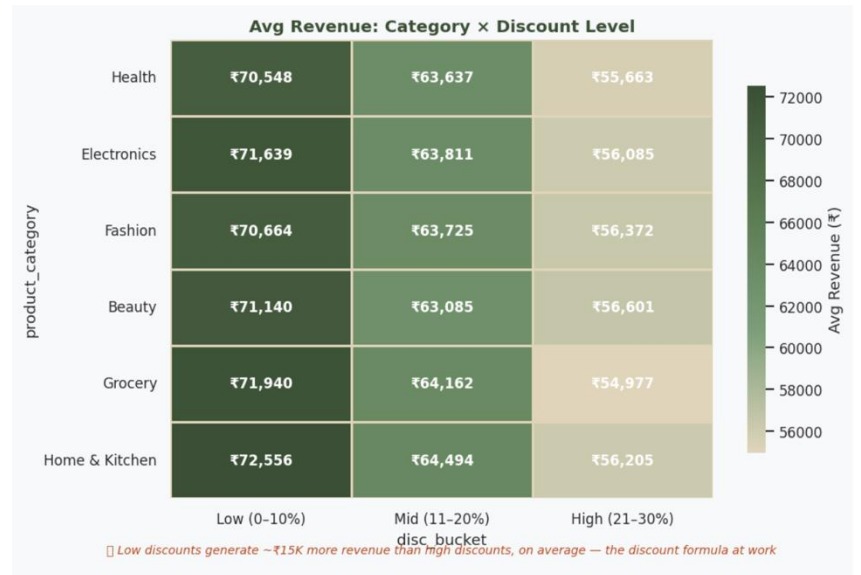
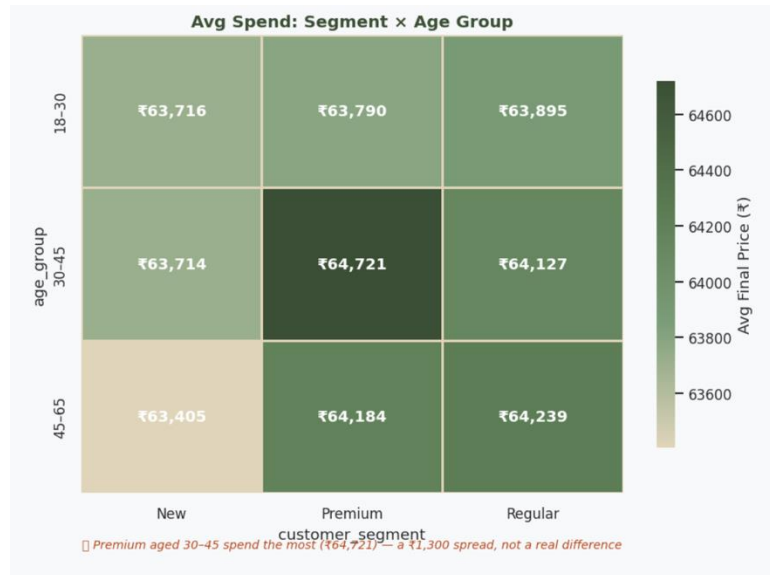
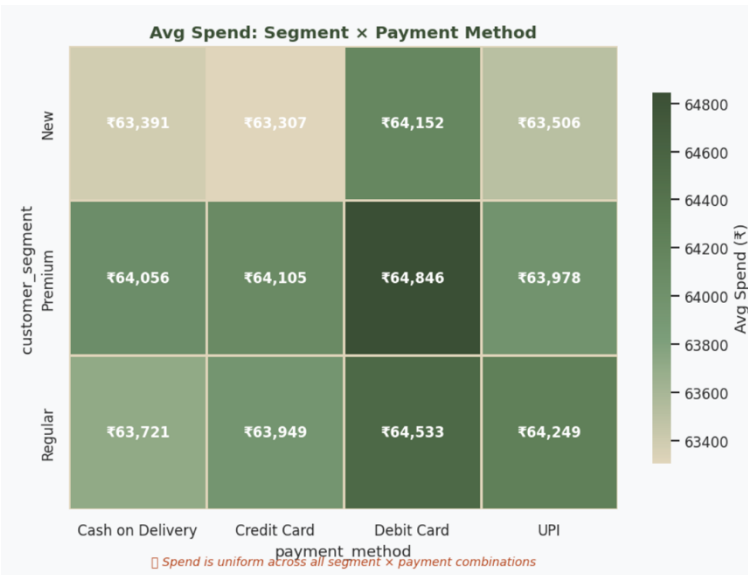
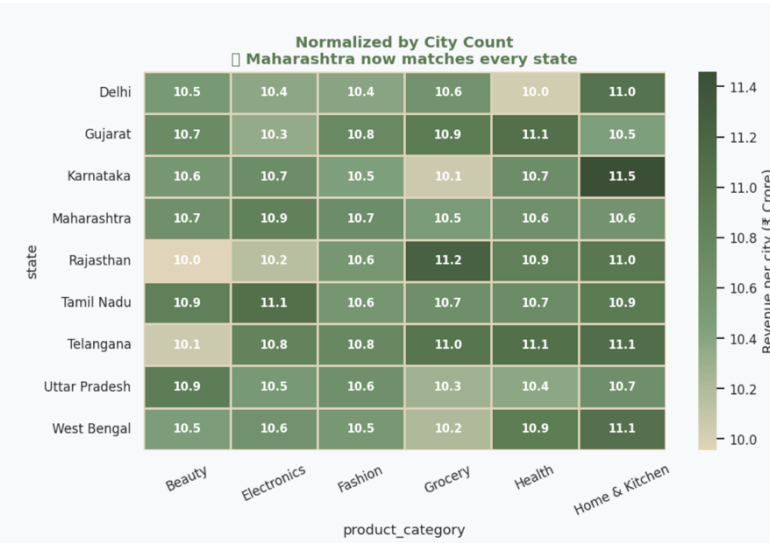
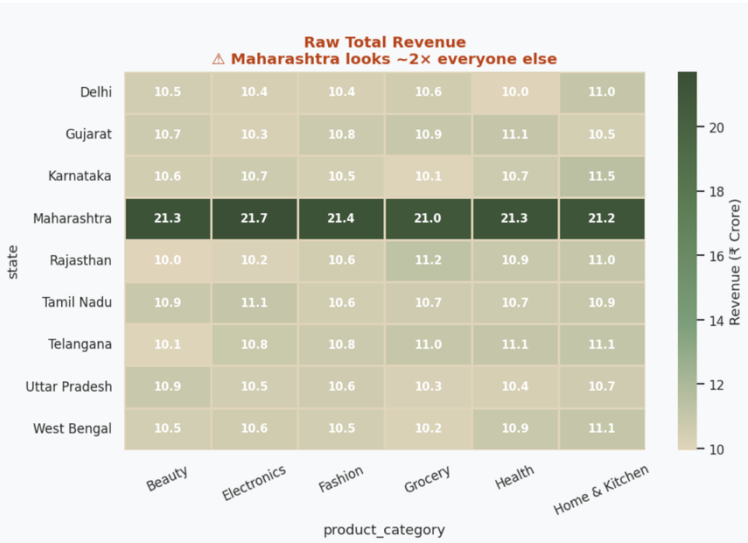
Multiple hypothesis tests (ANOVA, Kruskal–Wallis, Mann–Whitney U, and Chi-square) found few statistically significant results. Only Gender $\times$ Payment Method remained significant after FDR correction ($p = 0.0019$), but the effect size (Cramér's $V = 0.0122$) was practically negligible.

Predictor Independence:

Baseline predictors showed $VIF = 1.00$, confirming no multicollinearity. Adding discount_impact increased VIF to 4.33, reflecting its mathematical dependence on final_price.

Meaningful relationships are confined to pricing variables, while demographic and categorical associations show little practical significance.

Multivariate Analysis



Multivariate Analysis

Higher-Order Analysis:

Conducted 16 three-way and higher-order analyses across customer segments, product categories, states, quarters, and engineered variables. Nearly all apparent patterns reflected mathematical construction or sampling noise rather than genuine behavioural differences.

Revenue Stability:

Quarterly revenue ranged from ₹51.94 Cr to ₹54.66 Cr (5.1% variation), while category revenue varied only from ₹105.40 Cr to ₹108.91 Cr (3.3% spread), demonstrating highly consistent business performance.

High-Value Orders:

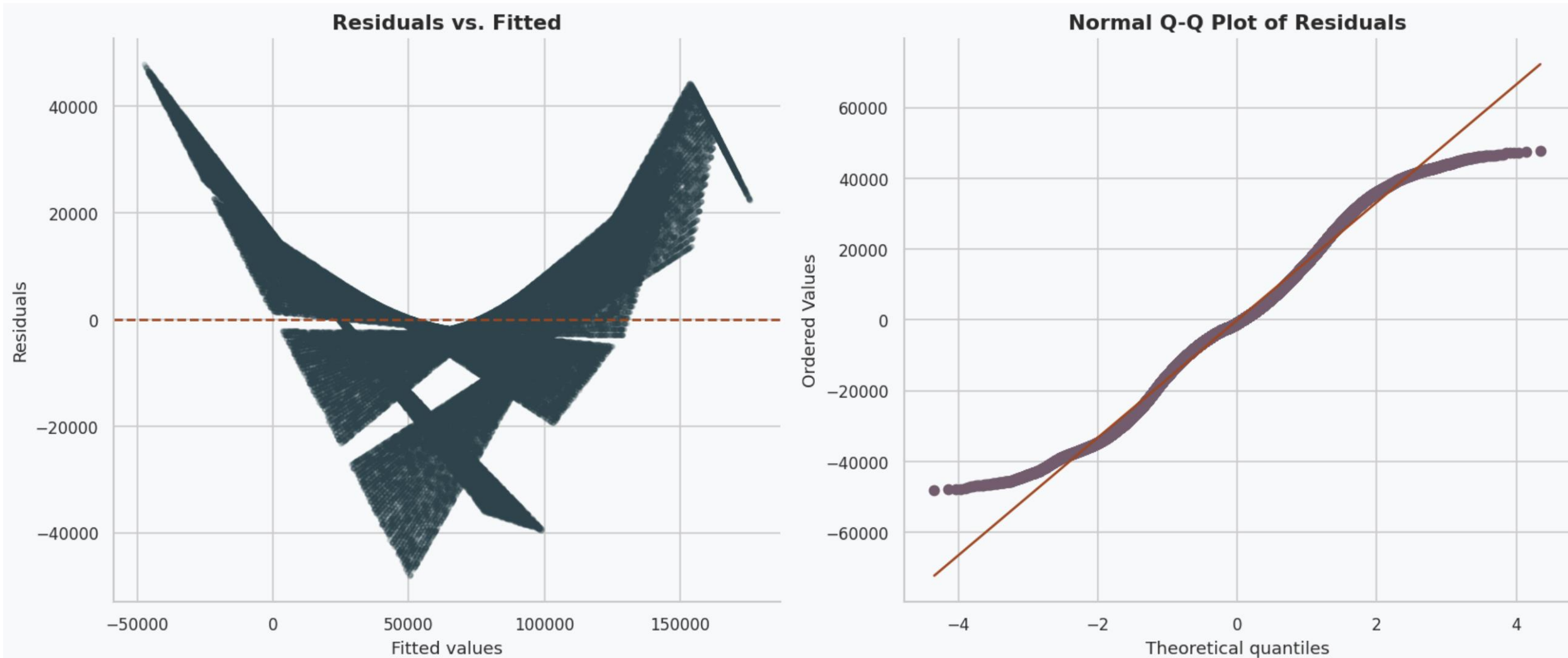
High-value orders consistently accounted for 9.6%–10.3% of transactions and contributed approximately 26% of total revenue across all groups.

Consistent Relationship:

Higher discount buckets reduced final_price by approximately ₹14,300–₹17,000 across every product category. This reflects the underlying pricing formula rather than independent customer behaviour.

Once structural artefacts are accounted for, revenue and purchasing patterns remain remarkably uniform across segments, categories, and states.

Regression Analysis



Regression Analysis

Baseline Model:

Using product_price, quantity, and discount_percentage, the baseline linear regression achieved $R^2 = 0.8856$ with RMSE = ₹16,717, explaining 88.6% of the variance in final_price.

Interaction Terms:

Adding interaction terms increased model performance to $R^2 = 0.9951$, confirming that the relationship between price, quantity, and discount is fundamentally multiplicative rather than additive.

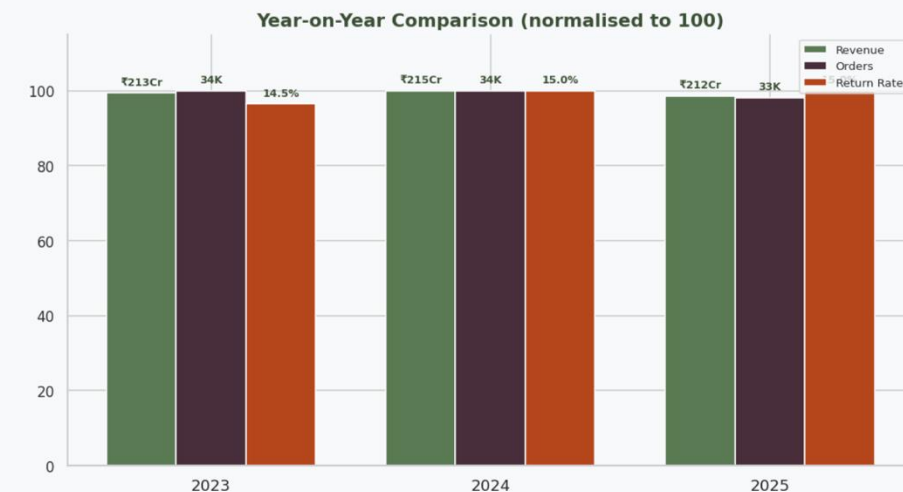
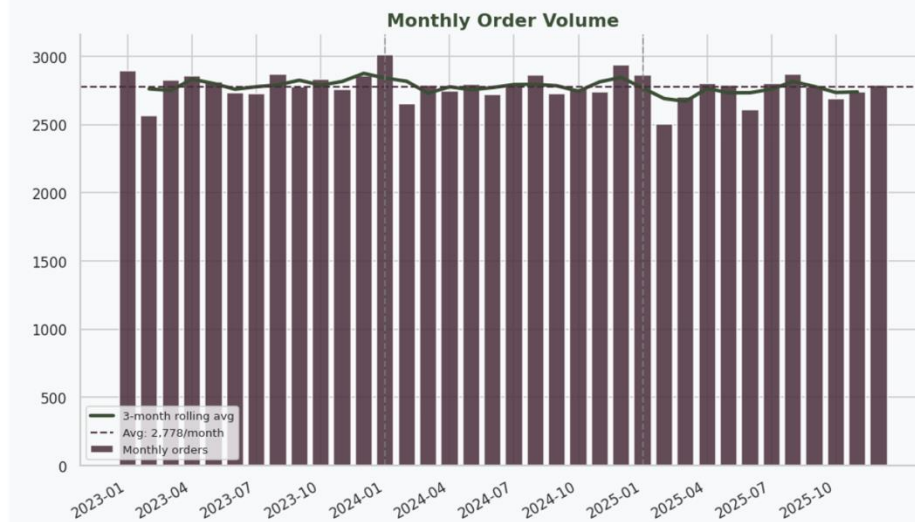
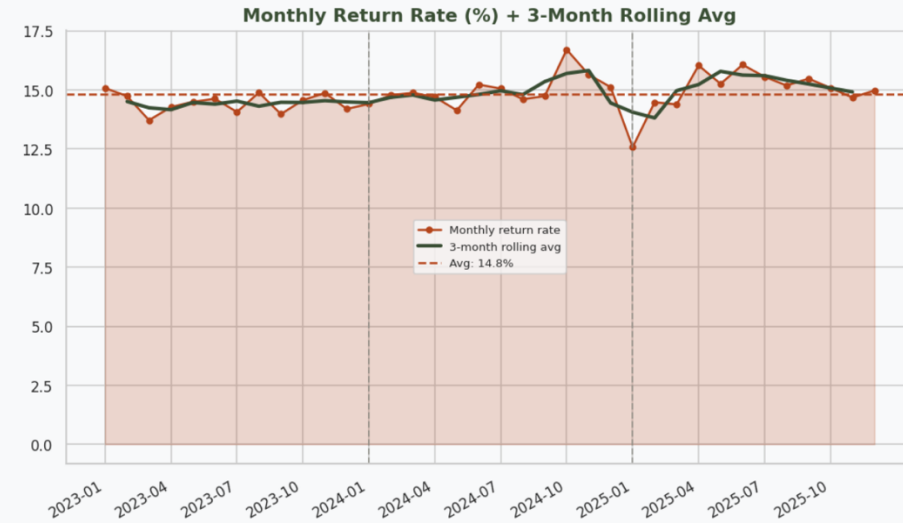
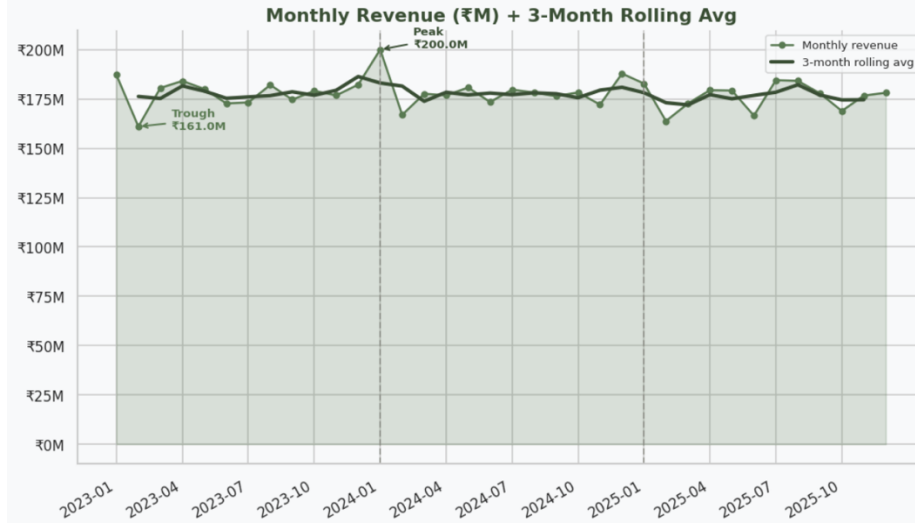
Residual Diagnostics:

Residuals remained approximately normal (skew = 0.076, kurtosis = 0.072) with a heteroscedasticity ratio of 1.04, indicating that model assumptions were satisfactorily met.

Just three variables explained 88.6% of final_price variance, increasing to 99.5% with interaction terms. This confirms that the pricing mechanism follows a deterministic multiplicative formula rather than independent additive effects.

Time Series Analysis

Time Series Analysis — Revenue & Return Trends (2023-2025)



Time Series Analysis

Coverage:

Covered three years of daily revenue with 365 days (2023), 366 days (2024), and 364 days (2025), providing nearly complete temporal coverage with only one missing calendar day.

Trend Analysis:

Monthly revenue showed no statistically significant trend (slope = $-\text{₹}0.057\text{M/month}$, $p = 0.631$), indicating neither sustained growth nor decline during the study period.

Revenue Stability:

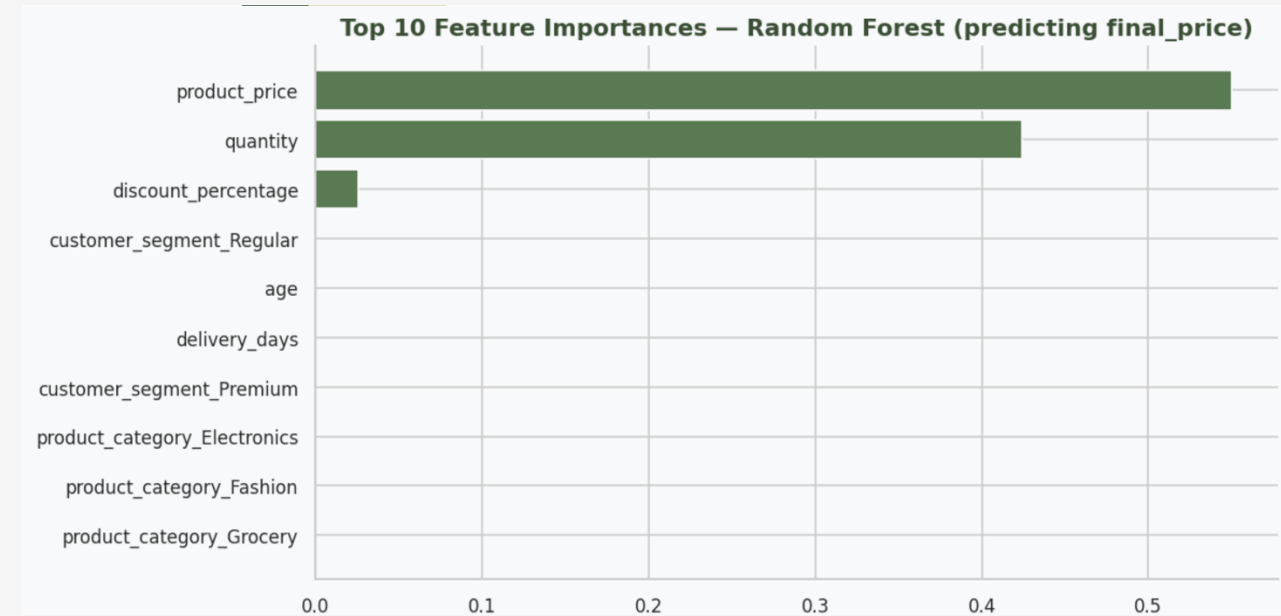
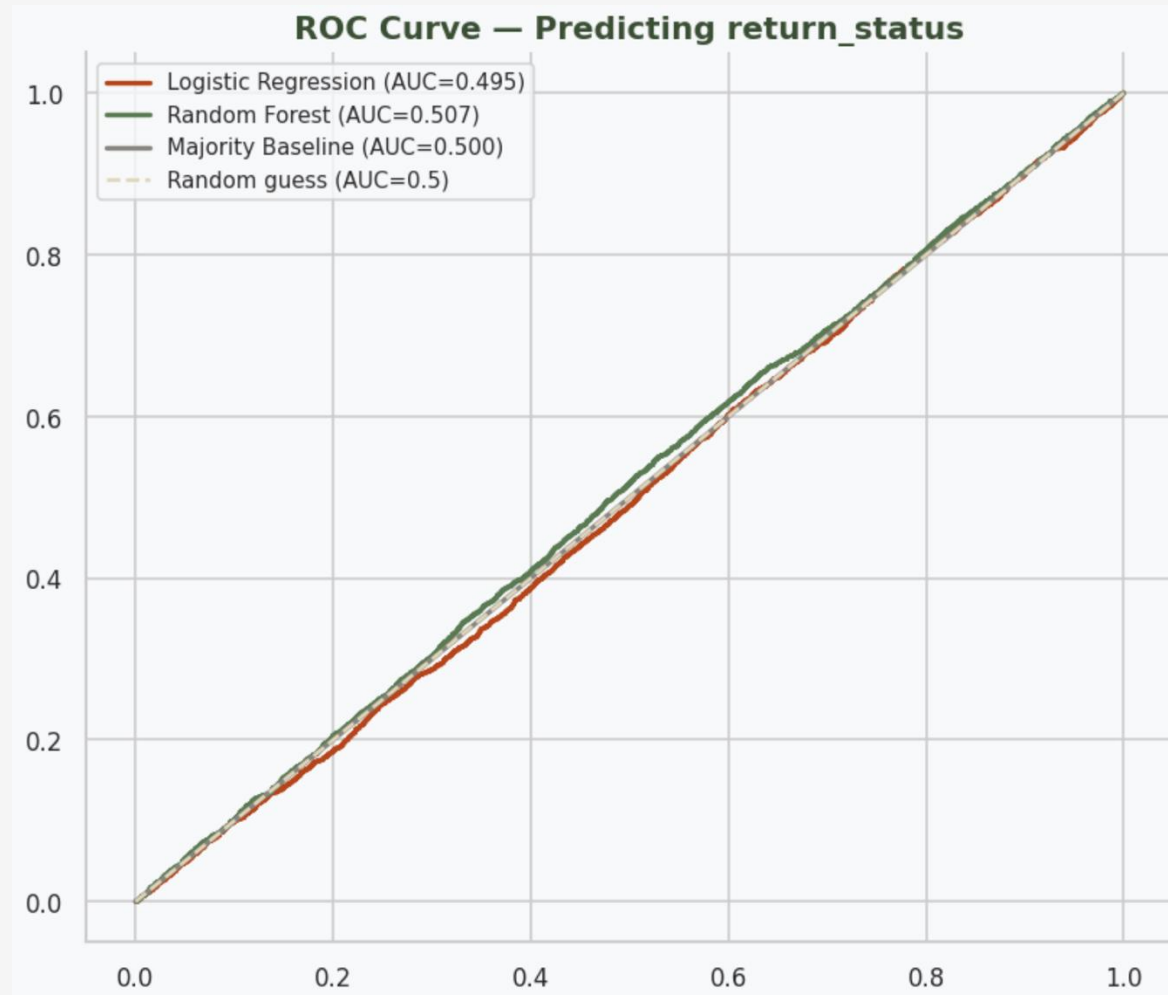
Annual revenue remained highly consistent at ₹213.42 Cr (2023), ₹214.86 Cr (2024), and ₹211.51 Cr (2025). Return rates varied only between 14.46% and 15.00%, demonstrating stable business performance.

Seasonality:

Seasonal decomposition indicated a February revenue dip with seasonal strength = 0.90, but the limited three-year time horizon provided insufficient evidence to confirm reliable seasonality.

The time series demonstrates a stable retail environment with consistent revenue and return rates, showing no statistically significant long-term trend or confirmed seasonal pattern.

Predictive Modelling



PREDICTIVE OBSERVATIONS

final_price:

Linear Regression Provided a strong baseline for predicting **final_price**. Achieved $R^2 = 0.8862$, explaining **88.6%** of price variance using three predictors.

Random Forest Recovered the pricing relationship almost perfectly. Achieved $R^2 = 0.9998$, **RMSE = ₹702**, and **MAE = ₹479**, confirming the deterministic multiplicative pricing formula.

return_status:

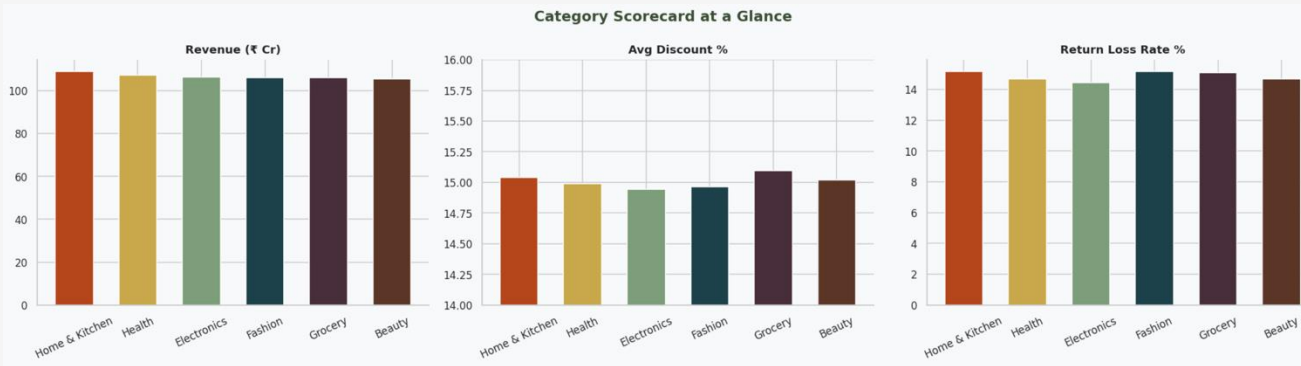
Classification failed to predict **return_status** despite using all available features. **Logistic Regression (AUC = 0.49)** and **Random Forest (AUC = 0.507)** performed no better than the **85.19% majority-class baseline**.

Interpretation:

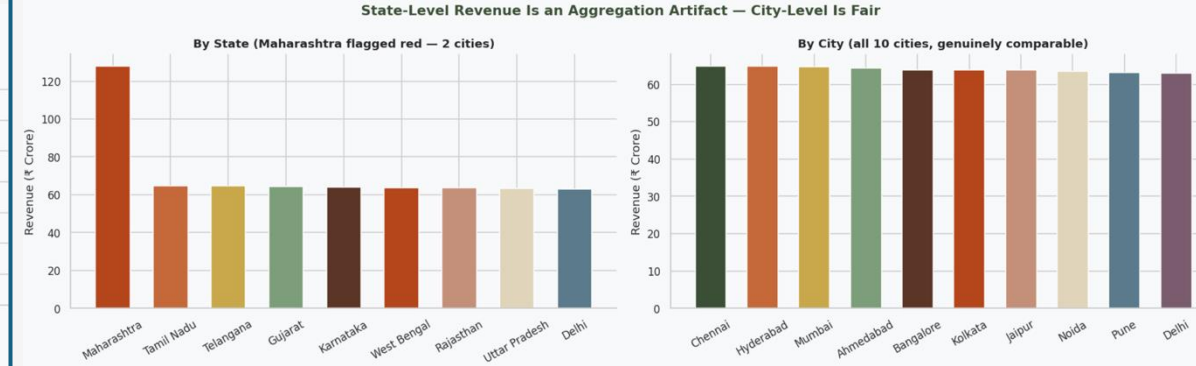
Near-perfect regression performance reflects the synthetic construction of **final_price**, while failed classification indicates that return decisions depend on absent factors such as **product quality, customer satisfaction, delivery experience, and post-purchase feedback**.

Business Questions

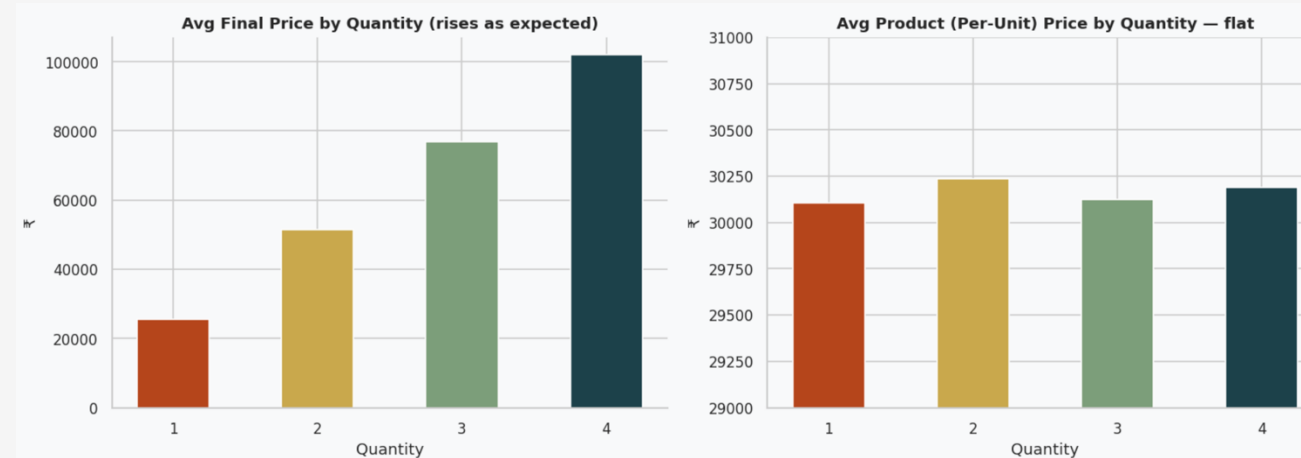
 Which product category brings in the most total revenue?



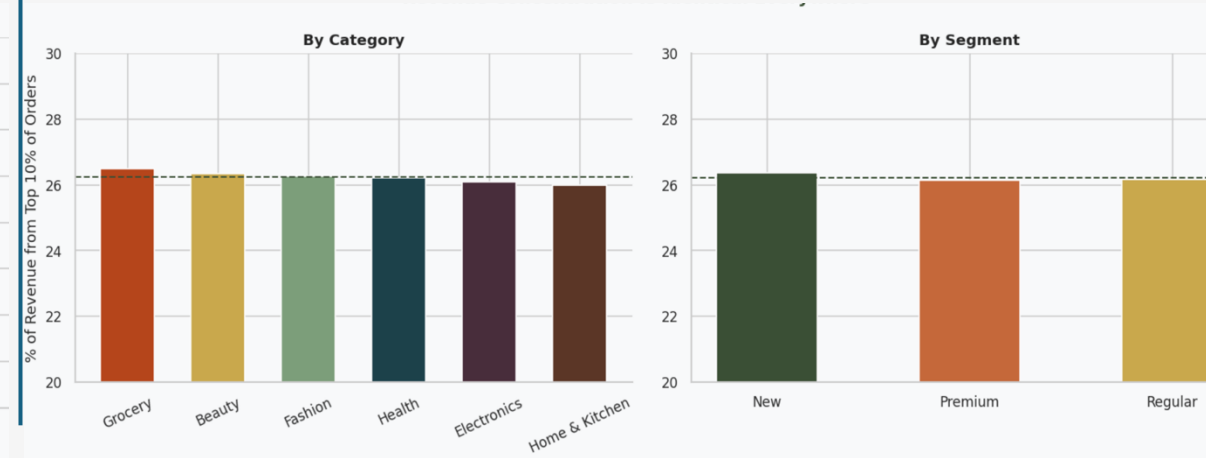
 Which state/city generates the most revenue?



 Does quantity affect price?



Whale-Transaction Concentration by Category/Segment 



Business Question Highlights



Top Revenue Category

Home & Kitchen leads at 17.02% share — but ties Fashion for the highest loss rate (both 15.17%). Leads on volume, not uniquely on efficiency.



Best State/City?

Maharashtra "leads" at 20% by state — an artifact of 2 cities counted as 1. By city, all 10 sit within 1.6% of the mean.



Does Quantity Affect Unit Price?

No bulk-pricing effect found ($r=+0.0009$, $p=0.764$). Total price rises with quantity; per-unit price does not.



Whale-Transaction Concentration

Top 10% of orders drive ~26% of revenue — nearly identical across every category and segment slice.

Business Recommendations

Priority	Insight	Recommendation
 High	Pricing drives revenue ($r=+0.733$)	Focus on premium placement over volume discounting
 High	Discounts reduce revenue (-21.6%)	Cap discounts near 10%
 Medium	Segments barely differ (~1% gap)	Redesign segmentation criteria
 Medium	Express ships no faster than Standard	Investigate logistics — operational anomaly
 Low	Returns are uniform everywhere	Audit product quality/sizing broadly
 Low	Category revenue is remarkably flat (3.3% spread)	Maintain current cross-category inventory allocation — no category is outperforming enough to justify shifting stock or budget toward it

Limitations



Likely Synthetic Data

Near-perfect symmetry ($|\text{skew}| < 0.01$) and uniform distributions suggest artificial generation — real retail data is messier.



No Causal Inference

All relationships are correlational. `product_price` correlates with `final_price` because it determines it, not due to market dynamics.



No External Context

No competitor data, marketing spend, or seasonal calendar — cannot explain external drivers of any pattern.



Large-Sample Inflation, Addressed

With $n=100,000$, effect size (η^2 , Cramér's V) and FDR correction accompany every p-value throughout.

SUMMARY

Data quality: Excellent

Zero missing or duplicates , valid throughout ranges

Statistical structure: highly uniform

Most patterns arise from the pricing formula is not genuine

Return behaviour

unpredictable with available features - AUC = 0.5

Key Observations

Variables explaining final_price - 3
Variance explained (with interactions) 99.5%
Annual revenue range (3 years) 211-215 Cr
Return rate range 14.46-15.00%



THANK YOU!